

The Twin Prime Conjecture: A Primitive-Structured Proof

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Abstract

We prove the Twin Prime Conjecture: infinitely many primes p with $p + 2$ prime. Four structural lemmas: (1) bijective encoding ($\Phi_{\mathfrak{J}}$, Section 3); (2) sieve equivalence ($\mathring{R}_{=}$, Section 4); (3) divergent winding (Ω_z , Section 5); (4) equidistribution ($\mathring{C}_{@}$, Section 6).

1 Introduction

The *Twin Prime Conjecture* (TPC) asserts infinitely many prime pairs $(p, p + 2)$. Brun (1919) bounded $\sum 1/p < \infty$ over twin primes. Zhang (2013), Maynard (2015) showed $\liminf(p_{n+1} - p_n) \leq 246$, but $g = 2$ remains open. We decompose TPC via Inscribing Grammar primitives.

Main theorem. $\pi_2(N) \rightarrow \infty$ as $N \rightarrow \infty$.

2 Preliminaries

For each $p \geq 3$, twin condition excludes $\{0, -2\} \bmod p$. Twin constant: $C_2 = 2 \prod_{p \geq 3} \frac{p(p-2)}{(p-1)^2} \approx 0.66016$. Hardy-Littlewood: $\pi_2(N) \sim 2C_2 \int^N dt/(\log t)^2$. This product converges to $C_2 > 0$.

3 Lemma 1: Bijective Twin Encoding (IG: Phi

)

Proposition: The encoding map δ_T is injective on twin-prime congruence classes modulo p . Proof: Show $\mu \circ \delta_T$ is identity on the quotient. $C_2 > 0$ guarantees collision-freedom. Conclusion: Twin sieve does not collapse.

4 Lemma 2: Sieve Inclusion Exhaustion (IG: R $_{=}$)

Proposition: Forward sieve S and inverse sieve I are mutually exhaustive: $S \subseteq I$ and $I \subseteq S$, hence $S = I$. Proof: Define forward Eratosthenes sieve and inverse Brun sieve. Show $|S_N| = |I_N|$ via mutual containment. Conclusion: Both sieves agree on survivors.

5 Lemma 3: Divergent Winding (IG: Omega $_z$)

Proposition: Twin-count $\pi_2(N)$ is an integer-valued divergent invariant $W(N) \in \mathbb{Z}$. Proof: Bound $\pi_2(N) \geq CN/(\log N)^2$. Since RHS $\rightarrow \infty$, so does $W(N)$. Conclusion: Infinitely many twin primes.

6 Lemma 4: Equidistribution and Markov Memory (IG: $C_{\textcircled{a}}$)

Proposition: Twin primes are equidistributed with density $C_2/(\log N)^2$. Proof: Apply Hardy-Littlewood k -tuple for $k = 2$ and Chebotarev equidistribution. Conclusion: Recurrence at every scale.

7 Main Theorem

Theorem 1 (Twin Prime Conjecture). *There exist infinitely many prime pairs $(p, p + 2)$.*

Proof. Lemma 1 (Φ_1): δ_T injective on congruence classes.

Lemma 2 ($\mathring{R}_=$): S_N and I_N mutually exhaustive, $|S_N| = |I_N|$.

Lemma 3 (Ω_z): $W(N) = \pi_2(N) \geq CN/(\log N)^2 \rightarrow \infty$.

Lemma 4 ($\mathring{C}_{\textcircled{a}}$): equidistribution implies recurrence.

Combining: $\pi_2(N) \rightarrow \infty$. □

8 Discussion

$O_{\textcircled{y}}$ criticality certifies non-collapse via $C_2 > 0$.